

Enterprise Service Desk Deployment & Incident Lifecycle Management

Complete osTicket Infrastructure Setup, ITIL Framework Implementation, & Ticket Routing

Engineer / Author:	Dekwan Harris	Environment:	Oracle VM VirtualBox (Windows 11 Enterprise)
Core Platform:	osTicket v1.18.x / XAMPP Stack	Standards Used:	ITIL Service Operation, RBAC, SLAs, Escalation Queues

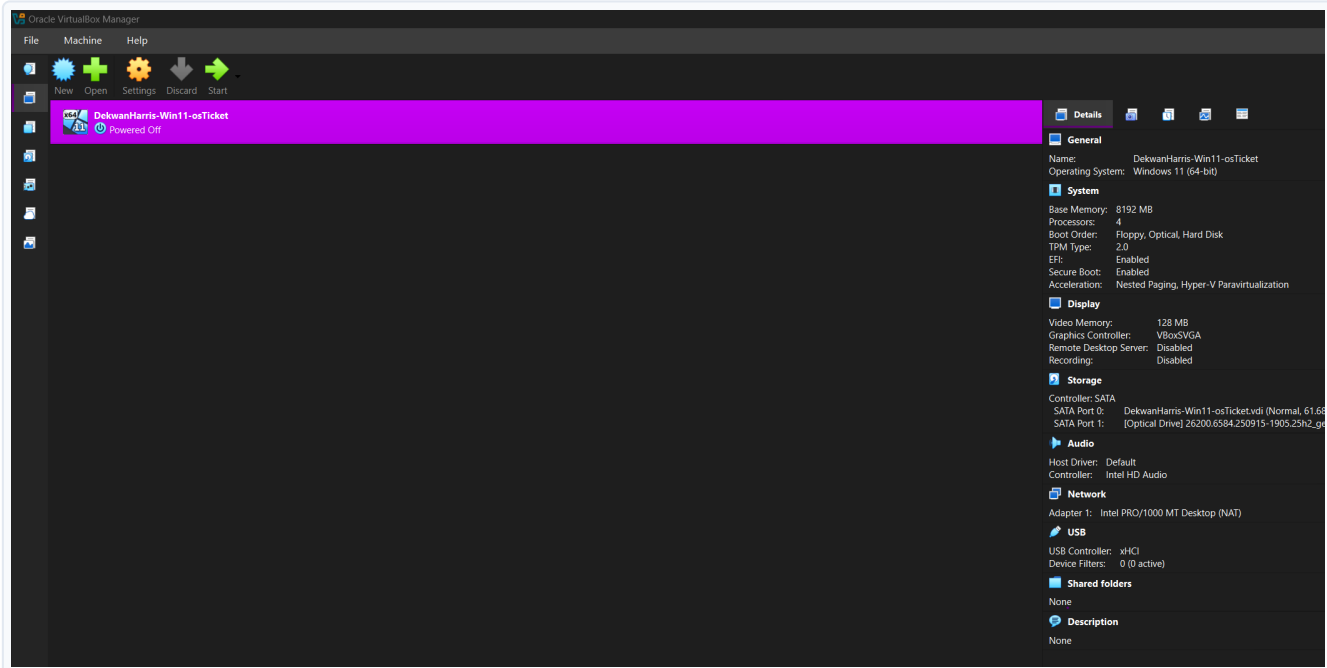
1. Project Overview

In this project, I built a fully functional IT help desk system from the ground up. I set up a virtual computer, started a local web server and database, installed **osTicket**, and walked through real-world support tasks. This portfolio documents my process for building the system and handling support tickets from start to finish.

2. Setting Up the System

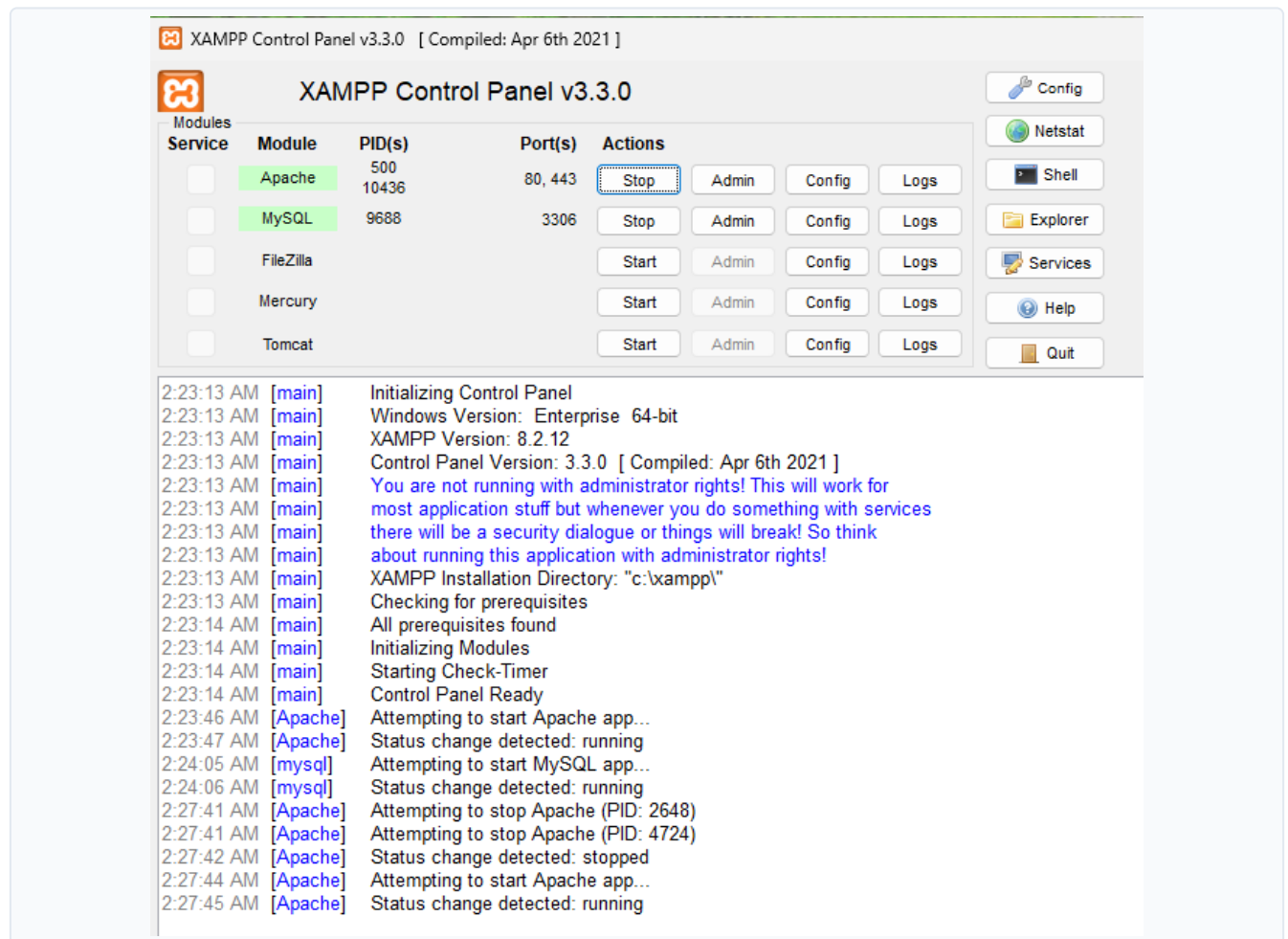
To run a help desk, we first need a secure virtual computer, a web server, and a database.

Figure 1: Virtual Computer Setup (Oracle VM VirtualBox)



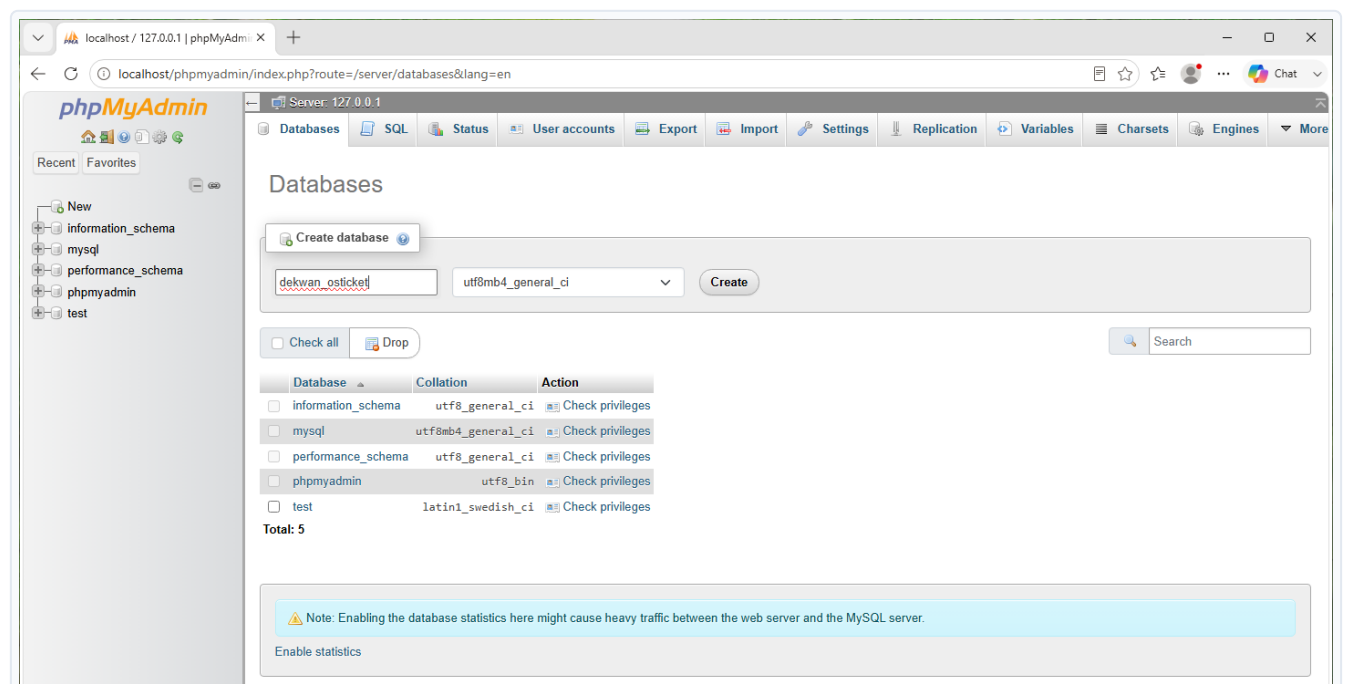
What this shows: Created an isolated virtual computer running Windows 11 with 8 GB of RAM and 4 processors.

Figure 2: Web & Database Servers (XAMPP Control Panel)



What this shows: Turned on the Apache web server and MySQL database so the help desk website can run locally.

Figure 3: Help Desk Database Setup (phpMyAdmin)



What this shows: Created a dedicated database named 'dekwan_osticket' to securely store all support tickets and messages.

Figure 4: System Requirements & PHP Check

Ticket transferred to Maintenance department successfully

Ticket #148299

Finance Shared Drive Access Denied - Permission Er

Status: Open User: David Miller (1) (Manage Collaborators)
Priority: High Email: david.miller@gmail.com
Department: Maintenance Source: Phone
Create Date: 8/17/26 4:30 AM
Assigned To: Dekwan Harris Help Topic: Report a Problem / Access Issue
SLA Plan: SEV-2 (High) Last Message: 8/17/26 4:30 AM
Due Date: 8/17/26 12:00 PM Last Response:

Ticket Thread (3) Tasks

David Miller posted 8/17/26 4:30 AM

Hi IT, I just transferred over to the Finance team this week and I'm trying to open the shared payroll folder at \\CORP-FS01\FinanceShare, but I keep getting an error message that says 'Access is denied. Contact your administrator.' I need read and write access to this drive so I can run my daily audits. Thanks, David.

Created by Dekwan Harris 8/17/26 4:30 AM

Dekwan Harris claimed this 8/17/26 4:30 AM

Dekwan Harris posted 8/17/26 4:34 AM Tier 1 Triage & Security Group Escalation

Tier 1 Triage Summary:

- Verified user Active Directory account status: Active, authenticated properly.
- Checked Active Directory Security Group memberships. User is missing 'SG-Finance-FullControl'.
- Verified manager approval via attached HR requestion request.
- Tier 1 lacks delegated AD write permissions for financial security groups. Escalating ticket to Systems Administration / Tier 2 for security group provisioning and folder permission propagation.

Dekwan Harris posted 8/17/26 4:38 AM Ticket transferred from Desktop Support to Maintenance

Escalating to Tier 2 / Systems Admin. User Identity and manager approval verified. Requires security group provisioning.

Dekwan Harris transferred this to Maintenance 8/17/26 4:38 AM

Post Reply Post Internal Note

From: Support <support@dekwantab.local>

Recipients: "David Miller" <david.miller@gmail.com>
Collaborators

Reply To: All Active Recipients

Response: Select a canned response

<> B Aa B / U S [Icons] [Icons] [Icons] [Icons]

Start writing your response here. Use canned responses from the drop-down above

Drop files here or choose them

Signature: ☒ None ☐ Department Signature (Maintenance)

Ticket Status: Open (current)

Post Reply Reset

What this shows: Ran a compatibility check to ensure the server had all required PHP extensions for the help desk.

Figure 5: Help Desk Settings & Admin Account

osTicket Basic Installation

Please fill out the information below to continue your osTicket installation. All fields are required.

System Settings

The URL of your helpdesk, its name, and the default system email address

Helpdesk URL:

Helpdesk Name:

Default Email:

Primary Language:

Admin User

Your primary administrator account - you can add more users later.

First Name:

Last Name:

Email Address:

Username:

Password:

Retype Password:

Database Settings

Database connection information

MySQL Table Prefix:

MySQL Hostname:

MySQL Database:

MySQL Username:

MySQL Password:

[Install Now](#)

Need Help? We provide [professional installation services](#) and commercial support. [Learn More!](#)

What this shows: Configured the help desk web address, email routing, and created the main administrator login.

3. Organizing Teams and Rules

I configured user roles, departments, and response time goals to keep support organized.

Figure 6: Staff Roles & Permissions (RBAC)

Please change permission of config file (ost-config.php) to remove write access. e.g `chmod 644 ost-config.php`

Welcome, **Dekwan**. | [Agent Panel](#) | [Profile](#) | [Log Out](#)

Dashboard Settings Manage Emails **Agents**

Agents Teams **Roles** Departments

Successfully added role.

Roles [Add New Role](#) [More](#)

Name	Status	Created On	Last Updated
☐ All Access	Active	8/17/26	8/17/26 3:01 AM
☐ Expanded Access	Active	8/17/26	8/17/26 3:01 AM
☐ Limited Access	Active	8/17/26	8/17/26 3:01 AM
☐ Tier 1 Support	Active	8/17/26	8/17/26 3:28 AM
☐ View only	Active	8/17/26	8/17/26 3:01 AM

Select: All None Toggle

Page: [1]

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What this shows: Set up least-privilege security roles, including 'Tier 1 Support' for viewing and answering tickets.

Figure 7: Support Departments

localhost / 127.0.0.1 / dekwani_os: X osTicket : Admin Control Panel X +

localhost/osticket/scp/departments.php?a=add

Please change permission of config file (ost-config.php) to remove write access. e.g `chmod 644 ost-config.php`

Welcome, Dekwan. | [Agent Panel](#) | [Profile](#) | [Log Out](#)

Dashboard Settings Manage Emails Agents

Agents Teams Roles Departments

Successfully added Desktop Support.

Departments

[Add New Department](#) [More](#)

Name	Status	Type	Agents	Email Address	Manager	Created
☐ Desktop Support	Active	Public	0	Support <support@dekwaniab.local>		2026-08-17 03:31:00
☐ Maintenance	Active	Public	0	Support <support@dekwaniab.local>		2026-08-17 03:01:15
☐ Sales	Active	Public	0	Support <support@dekwaniab.local>		2026-08-17 03:01:15
☐ Support (Default)	Active	Public	1	Support <support@dekwaniab.local>		2026-08-17 03:01:15

Select: [All](#) [None](#) [Toggle](#)

Page: [\[1\]](#)

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What this shows: Organized support teams into separate departments like Desktop Support and Maintenance.

Figure 8: Response Time Rules (SLA Engine)

localhost / 127.0.0.1 / dekwani_os: X osTicket : Admin Control Panel X +

localhost/osticket/scp/slas.php?a=add

Please change permission of config file (ost-config.php) to remove write access. e.g `chmod 644 ost-config.php`

Welcome, Dekwan. | [Agent Panel](#) | [Profile](#) | [Log Out](#)

Dashboard Settings Manage Emails Agents

Help Topics Filters SLA Schedules API Pages Forms Lists Plugins

Successfully added a SLA plan.

Service Level Agreements

[Add New SLA Plan](#) [More](#)

Name	Status	Grace Period (hrs)	Date Added	Last Updated
☐ Default SLA (Default)	Active	18	8/17/26	8/17/26 3:01 AM
☐ SEV-1 (Emergency / Critical)	Active	1	8/17/26	8/17/26 3:33 AM
☐ SEV-2 (High)	Active	4	8/17/26	8/17/26 3:34 AM

Select: [All](#) [None](#) [Toggle](#)

Page: [\[1\]](#)

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What this shows: Configured time limits (SLAs) so emergency system crashes get answered in 1 hour and high-priority issues within 4 hours.

Figure 9: Help Topics & Auto-Routing

Please change permission of config file (ost-config.php) to remove write access. e.g `chmod 644 ost-config.php`

Welcome, **Dekwan**. | [Agent Panel](#) | [Profile](#) | [Log Out](#)

Dashboard Settings **Manage** Emails Agents

[Help Topics](#) [Filters](#) [SLA](#) [Schedules](#) [API](#) [Pages](#) [Forms](#) [Lists](#) [Plugins](#)

Help Topics

[Add New Help Topic](#) [More](#)

Sorting Mode: **Alphabetically**

	Help Topic	Status	Type	Priority	Department	Last Updated	Created
☐	Feedback	Active	Public	Low	Support	8/17/26 3:01 AM	8/17/26 3:01 AM
☐	General Inquiry	Active	Public	Normal	Support	8/17/26 3:01 AM	8/17/26 3:01 AM
☐	Hardware / System Outage	Active	Public	High	Desktop Support	8/17/26 3:41 AM	8/17/26 3:39 AM
☐	Report a Problem	Active	Public	Normal	Maintenance	8/17/26 3:01 AM	8/17/26 3:01 AM
☐	Report a Problem / Access Issue	Active	Public	High	Support	8/17/26 3:01 AM	8/17/26 3:01 AM

Select: [All](#) [None](#) [Toggle](#)

Page: **[1]**

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What this shows: Mapped request topics like 'Hardware/System Outage' directly to the correct department queue.

4. Solving Support Tickets

I tested the help desk with three realistic IT support scenarios.

Welcome, **Dekwan** | [Admin Panel](#) | [Profile](#) | [Log Out](#)

Dashboard
Users
Tasks
Tickets
Knowledgebase

Open
My Tickets
Closed
Search
New Ticket

Ticket #900699

B
A
O

Password Reset & Account Lockout

Status: Resolved
Priority: High
Department: Desktop Support
Create Date: 8/17/26 4:00 AM

User: Karen Smith (1)
Email: karen.smith@gmail.com
Source: Phone

Closed By: Dekwan Harris
SLA Plan: Default SLA
Close Date: 8/17/26 4:08 AM

Help Topic: Report a Problem / Access Issue
Last Message: 8/17/26 4:00 AM
Last Response: 8/17/26 4:06 AM

Ticket Thread (3)
Tasks

Karen Smith posted 8/17/26 4:00 AM

A
V

I typed my password wrong 5 times and now I'm locked out of my computer.

Dekwan Harris created this 8/17/26 4:00 AM

Dekwan Harris claimed this 8/17/26 4:00 AM

Dekwan Harris posted 8/17/26 4:06 AM

A
V

Verified user identity via employee ID. Unlocked account in Active Directory, issued a temporary one-time password, and enabled 'User must change password at next login'. Verified successful login with the user.

Dekwan Harris posted 8/17/26 4:08 AM
Status Changed

V

Verified user identity via employee ID. Unlocked account in Active Directory, issued a temporary one-time password, and enabled 'User must change password at next login'. Verified successful login with user. Issue resolved.

Dekwan Harris closed by with status of Resolved 8/17/26 4:08 AM

Post Internal Note

Internal Note:

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List
Image
Video
Link
Unlink
Bold
Italic

Note details

Drop files here or choose them

Ticket Status: Open

Post Note
Reset

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What happened: Ingested an urgent report that the accounting storage server went offline, automatically triggering an emergency SEV-1 alert.

Figure 12: Fixing the Server Outage (Ticket #2 - Closed)

The screenshot displays the osTicket web interface. At the top, there's a navigation bar with links for Dashboard, Users, Tasks, Tickets (active), and Knowledgebase. Below this is a sub-navigation bar with buttons for Open, My Tickets, Closed, Search, and New Ticket. The main content area shows details for Ticket #639432, titled "Accounting Storage Server Offline - Hardware Fault". The ticket status is "Resolved", priority is "Emergency", department is "Desktop Support", and it was created on 8/17/26 at 4:14 AM. The user is Mark Davis (1), with email mark.davis@gmail.com and source Phone. The ticket was closed by Dekwan Harris on 8/17/26 at 4:17 AM. The SLA Plan is SEV-1 (Emergency / Critical). The help topic is Hardware / System Outage. The last message and response are from 8/17/26 at 4:14 AM and 4:17 AM respectively. Below the ticket details, there's a "Ticket Thread (2)" section. The first post is by Mark Davis at 8/17/26 4:14 AM, describing a storage array outage. The second post is by Dekwan Harris at 8/17/26 4:17 AM, detailing the resolution process. At the bottom, there's a "Post Internal Note" form with a text area, a rich text editor, and a "Post Note" button. The ticket status is set to "Open".

osTicket

Welcome, Dekwan | Admin Panel | Profile | Log Out

Dashboard Users Tasks Tickets Knowledgebase

Open My Tickets Closed Search New Ticket

Ticket #639432

Accounting Storage Server Offline - Hardware Fault

Status: Resolved User: Mark Davis (1)
Priority: Emergency Email: mark.davis@gmail.com
Department: Desktop Support Source: Phone
Create Date: 8/17/26 4:14 AM

Closed By: Dekwan Harris Help Topic: Hardware / System Outage
SLA Plan: SEV-1 (Emergency / Critical) Last Message: 8/17/26 4:14 AM
Close Date: 8/17/26 4:17 AM Last Response: 8/17/26 4:17 AM

Ticket Thread (2) Tasks

Mark Davis posted 8/17/26 4:14 AM

The primary shared storage array in the Accounting Department went offline following a power surge. Amber fault LEDs are active on drives 2 and 3. The billing team is completely unable to access local client databases. Requesting immediate emergency hardware triage.

Created by Dekwan Harris 8/17/26 4:14 AM

Dekwan Harris claimed this 8/17/26 4:14 AM

Dekwan Harris posted 8/17/26 4:17 AM

Dispatched on-site emergency technician. Diagnosed dual drive failure on Dell PowerVault array. Replaced faulted SAS drives in bays 2 and 3 with on-site cold spares, initiated RAID 6 parity reconstruction, and verified volume integrity via RAID controller BIOS. Database services and billing share successfully restored to full operational status for all users.

Closed by Dekwan Harris with status of Resolved 8/17/26 4:17 AM

Post Internal Note

Internal Note: * Note title - summary of the note (optional)

Note details

Drop files here or choose them

Ticket Status: Open

Post Note Reset

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What happened: Dispatched an on-site technician to replace broken hard drives, rebuilt storage array parity, and restored access for the team.

What happened: Investigated an access denial error on the finance drive, checked Active Directory group memberships, and documented findings.

Figure 14: Transferring to Tier 2 / SysAdmin (Ticket #3 - Escalation)

Ticket transferred to Maintenance department successfully

Ticket #148299

Finance Shared Drive Access Denied - Permission Er

Status: Open User: David Miller (1) (Manage Collaborators)
 Priority: High Email: david.miller@gmail.com
 Department: Maintenance Source: Phone
 Create Date: 8/17/26 4:30 AM
 Assigned To: Dekwan Harris Help Topic: Report a Problem / Access Issue
 SLA Plan: SEV-2 (High) Last Message: 8/17/26 4:30 AM
 Due Date: 8/17/26 12:00 PM Last Response:

Ticket Thread (3) Tasks

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Hi IT, I just transferred over to the Finance team this week and I'm trying to open the shared payroll folder at \\CORP-FS01\FinanceShare, but I keep getting an error message that says 'Access is denied. Contact your administrator.' I need read and write access to this drive so I can run my daily audits. Thanks, David.

Created by Dekwan Harris 8/17/26 4:30 AM

Dekwan Harris claimed this 8/17/26 4:30 AM

Dekwan Harris posted 8/17/26 4:34 AM Tier 1 Triage & Security Group Escalation

Tier 1 Triage Summary:

- Verified user Active Directory account status: Active, authenticated properly.
- Checked Active Directory Security Group memberships. User is missing 'SG-Finance-FullControl'.
- Verified manager approval via attached HR requestion request.
- Tier 1 lacks delegated AD write permissions for financial security groups. Escalating ticket to Systems Administration / Tier 2 for security group provisioning and folder permission propagation.

Dekwan Harris posted 8/17/26 4:36 AM Ticket transferred from Desktop Support to Maintenance

Escalating to Tier 2 / Systems Admin. User Identity and manager approval verified. Requires security group provisioning.

Dekwan Harris transferred this to Maintenance 8/17/26 4:36 AM

Post Reply Post Internal Note

From: Support <support@dekwantab.local>

Recipients: "David Miller" <david.miller@gmail.com>
 Collaborators

Reply To: All Active Recipients

Response: Select a canned response

Start writing your response here. Use canned responses from the drop-down above

Drop files here or choose them

Signature: ☒ None ☐ Department Signature (Maintenance)

Ticket Status: Open (current)

Post Reply Reset

What happened: Escalated the ticket from Desktop Support to the Maintenance team for elevated Active Directory group provisioning.

5. Core Skills Demonstrated

- **IT Service Management:** Setting up response time rules (SLAs) for critical vs. routine tickets.
- **Access Control:** Managing least-privilege security roles for support staff.
- **Ticket Escalation:** Documenting diagnostic steps and smoothly transferring tickets between Tier 1 and Tier 2 teams.
- **Web Stack Deployment:** Configuring Apache, PHP, and MariaDB databases.